



VOLTEGE SWAP & RIDE ECO-SYSTEM

EXPERIENCE THE FUTURE OF DRIVING TODAY



Presented by:
Low Risk, Lower Effort

SITUATION ANALYSIS

KEY MARKET FACTS

3%

Current EV penetration in Bangladesh (mainly 3-wheelers)

70%

EVs are unregulated e-rickshaws in peri-urban & rural areas

\$0.08/km

Avg EV running cost (70% cheaper than petrol)

90%

Charging done via unsafe home connections

MARKET GAPS



No Formal Charging Infra

Only 40 legal charging points nationwide



Policy Ambiguity

No dedicated EV tariff or BaaS regulation



Limited Local Battery Production

High import dependence

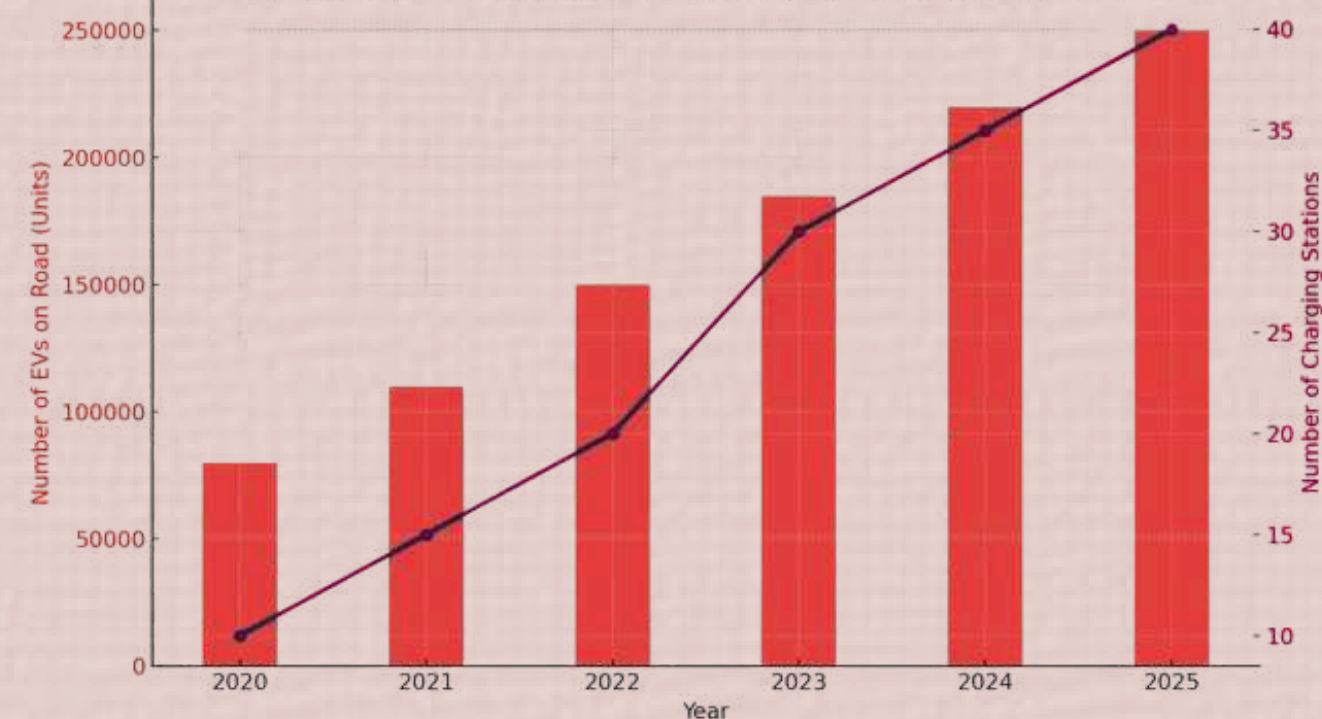


Opportunity

\$2B EV market by 2030 if infrastructure grows

EV adoption is rising fast, but infrastructure and policy are lagging behind—VoltEdge bridges this gap

EV Adoption vs Charging Stations in Bangladesh (2020-2025)



Source References for Credibility

BRTA Annual Transport Report 2024

SREDA EV Market Study 2025

IDCOL EV Financing Program

PROBLEM IDENTIFICATION

Bangladesh's electric vehicle adoption is accelerating, but the charging infrastructure, policies, and consumer readiness are not keeping pace—creating a critical gap that threatens sustainable growth



Core Issue



Impact Statement

> Without urgent infrastructure expansion and clear policies, EV growth will stall, and Bangladesh will struggle to achieve its clean energy transition goals.



Limited Charging Infrastructure

Only ~40 legal charging stations nationwide, mostly in Dhaka & Chattogram

Insufficient for 2,50,000+ EVs on the road today



Unsafe & Informal Charging

90% EVs depend on home or roadside connections

Causes fire hazards, power losses, and battery degradation



Policy & Regulatory Gaps

No dedicated EV tariff, Battery-as-a-Service (BaaS) regulation, or land allocation for hubs

Lack of standardization deters private investment



Consumer Adoption Barriers

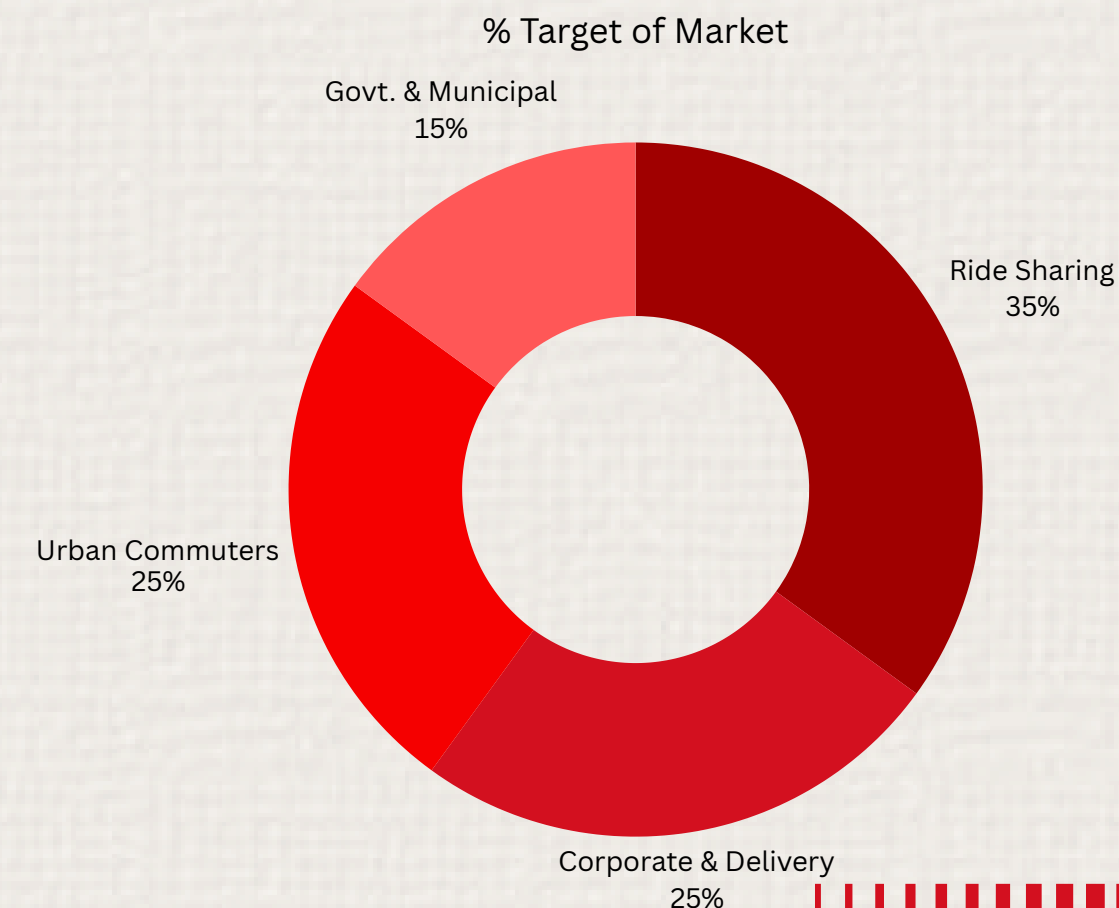
Range anxiety and battery life fears limit confidence

Low public awareness slows mainstream adoption



TARGET MARKET ANALYSIS

01 Primary Early Adopter Segments



Segment	Characteristics	Why They Adopt Early
Ride-Sharing Drivers (Uber, Pathao, Obhai)	High daily mileage, cost-conscious	Lower fuel cost, higher daily savings
Corporate & Delivery Fleets (e.g., bKash, Daraz, Pathao Courier)	Large vehicle pools, eco-conscious brands	Bulk cost reduction & ESG branding
Urban Commuters (Middle Class)	City-based, short-distance travel	Range fits city needs, eco-friendly image
Government & Municipal Vehicles	Policy-driven adoption	Low operating cost, showcase green leadership

02 Market Size & Opportunity

EV Market (2025): ~2,50,000 vehicles (mostly 2 & 3-wheelers)

Target Reach in 3 Years:

Ride-sharing & delivery fleets: 30% of market

Urban commuters & corporate fleets: 20% of market

TAM (Total Addressable Market): ~3,50,000 potential EV users by 2027



SURVEY SPEAKS

- > Primary daily transport is 2-3 wheelers **45%**
- > Have you used/experienced an EV? **35% YES**
- > Biggest concern → Charging availability **50%**
- > Prefer battery swapping over plug-in **70% YES**
- > Likely to use a Super App (rating $\geq 4/5$) **75%**
- > CO₂ savings motivates EV adoption **85% YES**

Rickshaw Drivers & Riders

1. "If battery swap takes less than 5 minutes, I'll switch to EV tomorrow." – CNG driver, Motijheel

2. "Paying monthly for battery is easier than buying a new one." – Rickshaw puller, Chawkbazar

"I'd try an EV if I could swap and ride without worrying about charging. And A Super App with routes and discounts will make me switch faster" – DU student, TSC



BIG IDEA: SWAP & RIDE - VOLTEGE EV ECOSYSTEM



Modular Solar-Powered Charging & Swapping Hubs

Grid + solar micro-hubs for 2, 3 & 4-wheelers
Battery-as-a-Service (BaaS) → Lower upfront cost
Fast-charging lanes for cars & fleet vehicles



VoltEdge Super App

Real-time charging & swapping map
Smart route planning & fleet management
Subscriptions, mobile payments & loyalty points



Public-Private Partnership (PPP) Model

Govt. → Land & policy support
VoltEdge + IDCOL + Ride-sharing apps → Invest & operate
Scalable, modular rollout for feasible budgets

Expected Impact by 2028

Metric	2025	2028 (Goal)
Public Charging Stations	40	500+ Hubs
EV Fleet on Road	2,50,000	6,00,000+
EVs Using Safe Charging	10%	70%+
CO ₂ Reduction/Year	0.05 Mt	0.3 Mt

Source: SREDA EV Roadmap 2022, IDCOL EV Program 2024

Build Bangladesh's first Swap & Ride EV ecosystem with solar-powered modular hubs, battery-swapping, and a digital fleet network to drive mass adoption

ACTION PLAN

01 Launch (2025-2026)

50 Modular Hubs in Dhaka & Chattogram

VoltEdge Super App launch → live map, swapping, mobile payment

Early Incentives: 20-30% cheaper Battery-as-a-Service (2 & 3-wheelers)

KPI: 10,000+ EVs onboard

02 Grow & Engage (2027-2028)

+200 Hubs along metro & highway corridors (Dhaka-Ctg, Dhaka-Sylhet, Dhaka-Khulna)

App Upgrades:

Smart route planning & predictive charging

Fleet dashboards for Uber/Pathao/Shohoz

Gamified loyalty → more rides, more swaps

Impact: 3,50,000+ EVs | CO₂ cut 0.15 Mt/year

03 Scale Nationwide (2029-2030)

500+ Hubs covering all divisional cities & highways

App Evolution:

AI-based swap booking & renewable energy optimization

Carbon credit tracking for govt & fleets

Impact: 6,00,000+ EVs | 70% safe charging | 0.3 Mt CO₂ cut/year

04 By 2030

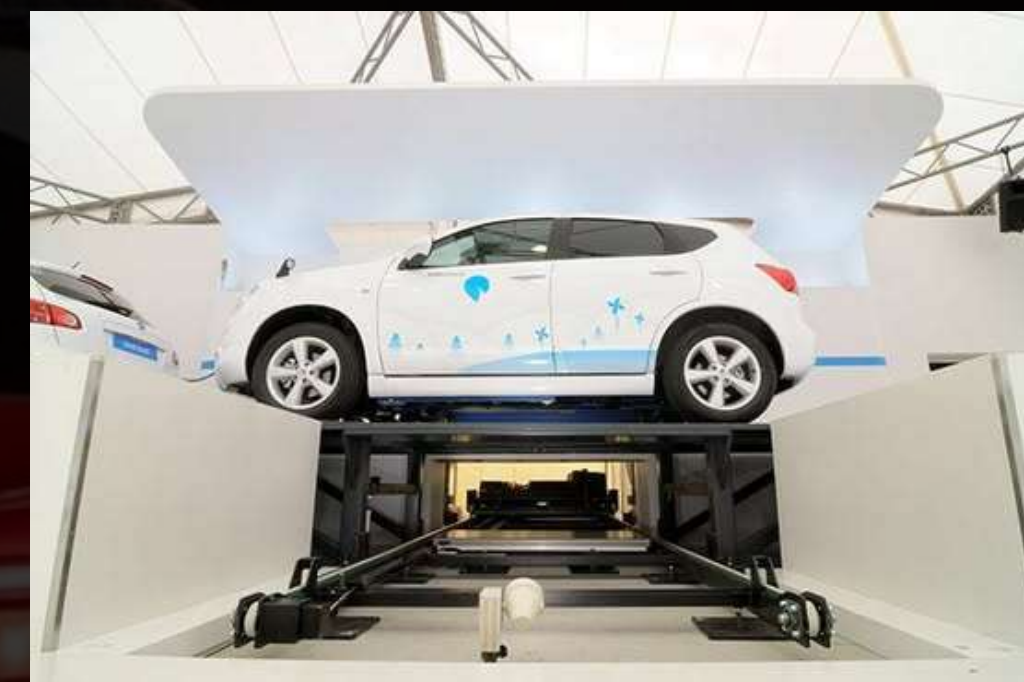
EVs Everywhere → 500+ hubs, nationwide coverage

VoltEdge Super App → 1-stop digital ecosystem

CO₂ Reduction → 0.3 Mt/year, supports 2030 Clean Energy Target

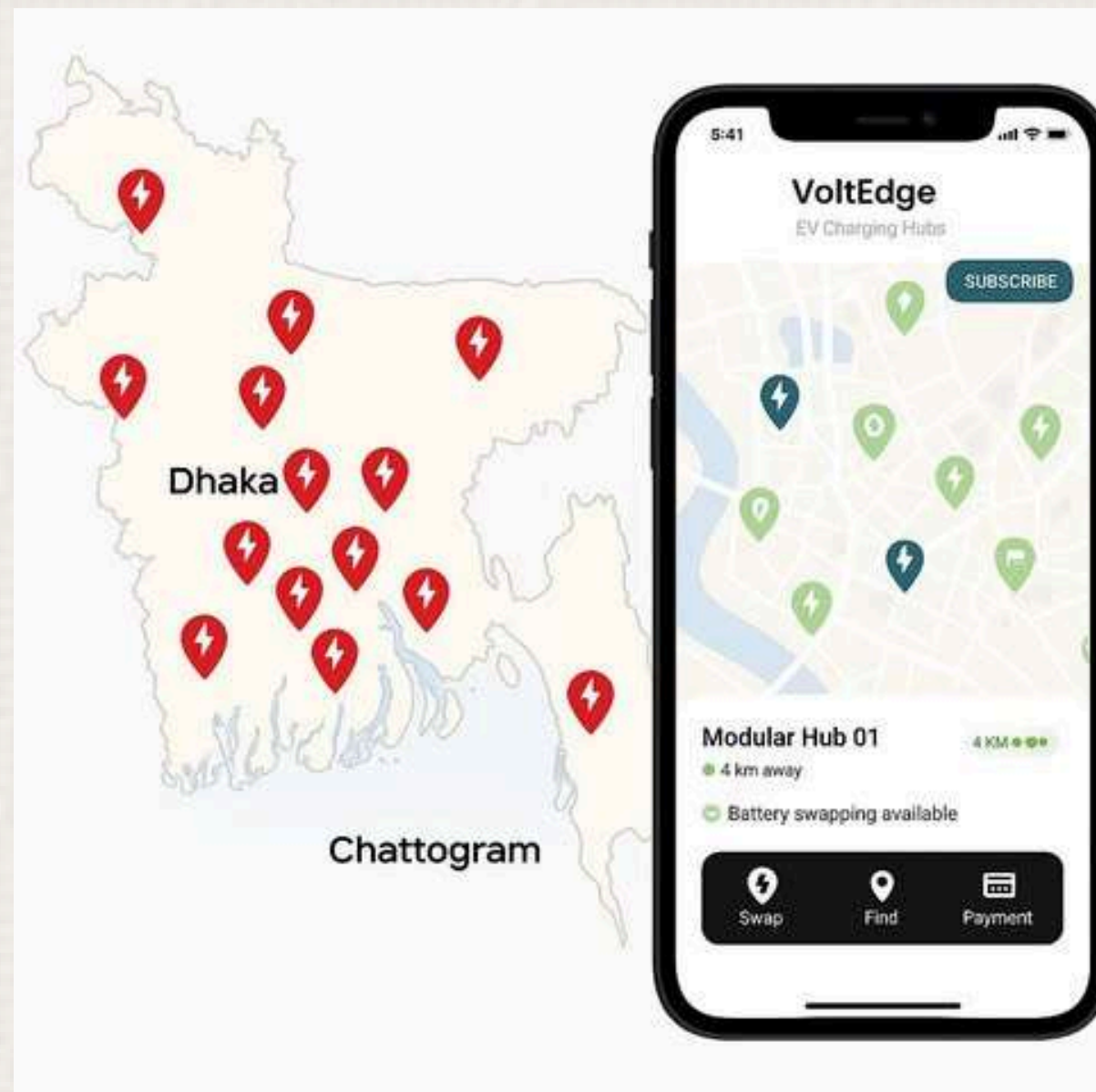
Data Sources:

SREDA EV Roadmap 2022 | IDCOL EV Program 2024 | IEA Global EV Outlook 2024



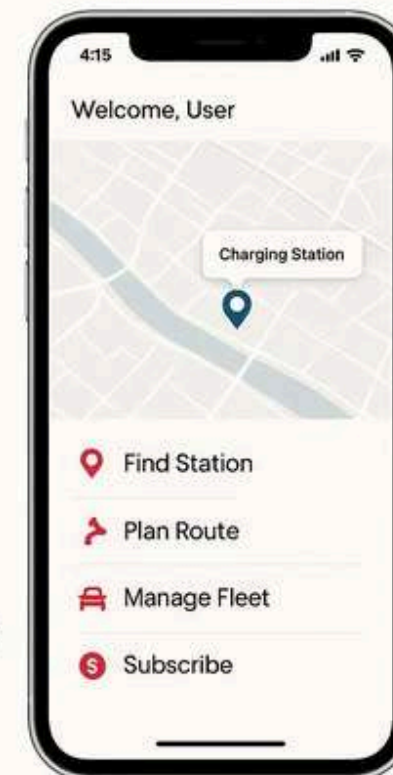
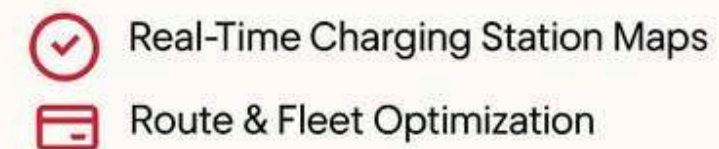
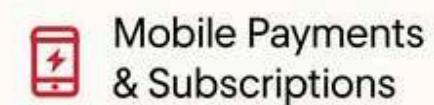
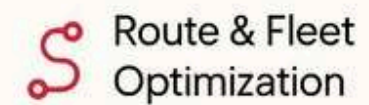
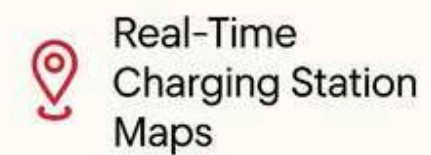


INTRODUCING **VOLTEGE** **SUPER APP**



SUPER APP

A digital platform to support EV users with smart charging solutions



Fleet Dashboard



Deliveries
1,250



Fleet EVs
280



Points
12,500



Fleet

Uber Pathao

Home Stations Fleet Alerts



VOLTEDGE EV ROADMAP

PHASE 1: Launch (2025–2026)

50 modular charging & swap hubs in Dhaka & Chattogram
10,000 EVs onboard (focus: 2 & 3-wheelers)
VoltEdge Super App launch (Live map + mobile payment)
KPI Bubble: 1.6 MW → 10K EVs

1

2

PHASE 2: Grow (2027–2028)

+200 hubs along Dhaka–Ctg, Sylhet, Khulna corridors
App upgrade: smart route planning + fleet dashboards
Fleet adoption: Uber, Pathao, Shohoz → 3,50,000 EVs
Impact Icon: 0.15 Mt CO₂ cut/year

3

PHASE 3: Scale Nationwide (2029–2030)

500+ hubs, nationwide coverage including all divisional cities
AI swap booking + renewable energy optimization in app
Carbon credit & ESG reporting for partners
Impact Badge: 0.3 Mt CO₂ cut/year | 70% safe charging

Opportunities
Don't Happen
You Create Them



VOLTEGE **POLICY** ***& REGULATORY*** ***SUPPORT***

CO₂ Reduction (Environmental)

Each battery swap saves 0.8 kg CO₂

5-Year Projection: ~12,000 tons CO₂ avoided

Equivalent: 600,000 trees planted

Job Creation (Social)

Pilot Phase: 120 local jobs (hub operators, maintenance)

Expansion: 1,000+ jobs by 2028

Special Focus: Rural youth employment

Women in EV Workforce (Inclusion)

Target: 30% female staff in app ops & customer support

Training: EV maintenance workshops for women

Impact: Promote gender-inclusive green mobility

Carbon Credit & Tree Planting (ESG)

Swap = Carbon Credit (offsets for fleets & corporates)

CSR Program: 1 tree planted for every 100 swaps

Brand Impact: Strong ESG alignment with SDG 7 & 13

SOCIAL & ENVIRONMENTAL IMPACT

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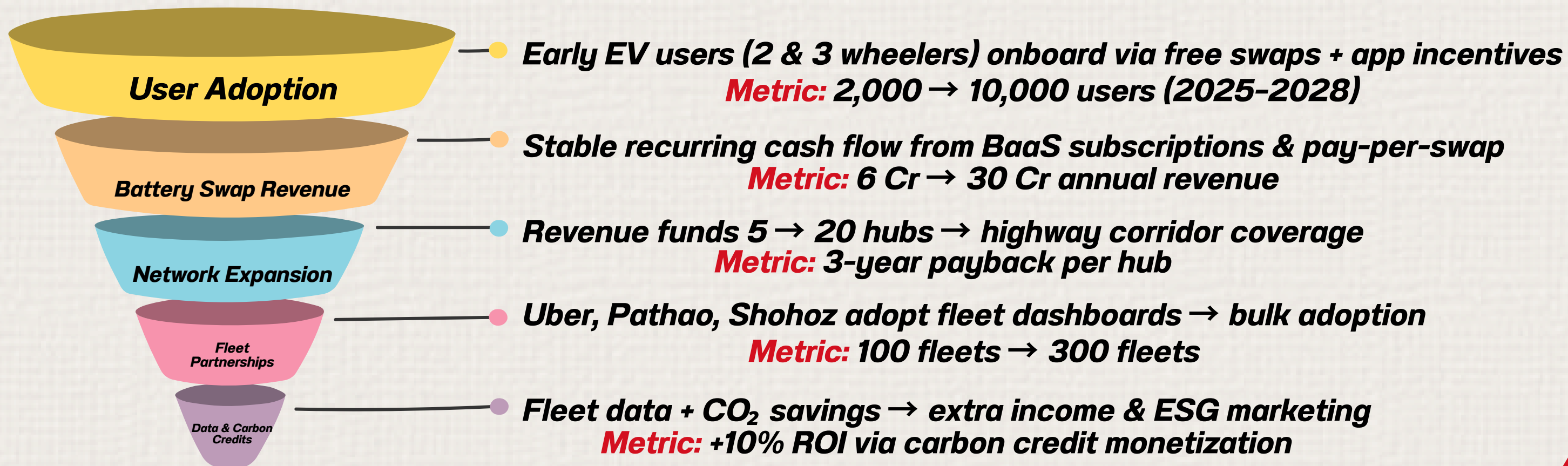
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VOLTEGE ECOSYSTEM FLYWHEEL





VOLT**EDGE** MARKETING PLAN: “**SWAP. RIDE. BELONG**”



Strategy

Instagram Reels War
VoltEdge Heroes Club
Swap Carnival

Unique Execution

1-minute battery swap challenge with influencers – fastest swap wins free BaaS for 6 months
First 100 riders become “Founding Swappers” with badges, exclusive jackets, leaderboard in app
Battery-swap tents in Shahbagh & GEC Circle with live music, test rides, tree giveaway



GROWTH: “Your Commute, Recharged” (2027–2028)



Strategy

App Loyalty Game
Branded Riders
Metro Wrap Ads
In-App Story Feed Share “My EV Story”

Unique Execution

Swaps earn “GreenPoints” → unlock badges, ride discounts, trees planted in your name
500 Pathao/Uber riders wear VoltEdge jackets + QR on helmets for referral cashback
Wrap Dhaka metro rakes with “Swap in 60s” & “Ride Without Smoke” visuals
Most inspiring wins media spotlight & free swaps for 1 year

RETENTION: “Ride Green, Live Clean”



Strategy

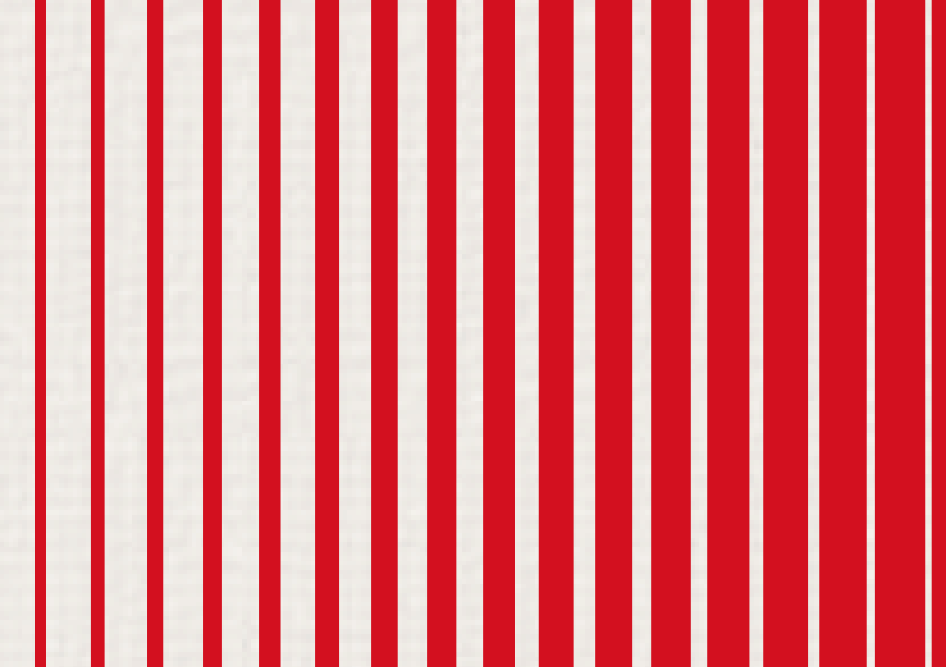
CO₂ Tracker Badge
Swap to Plant
“VoltEdge Rider” Day
Community Referral Net

Unique Execution

Real-time tracker in app: “You saved 0.4 tons CO₂. You're greener than 80% of riders!”
Every 100 swaps = 1 tree planted → GPS-verified photo sent to user
Monthly meetup of top riders at hubs → share stories, rewards, recognition
Riders refer, form ‘charging circles’ → top circles win VoltEdge “Green Gang” trophies



VOLTEGE PILOT BUDGET



Category	Unit Cost (BDT)	Scale	Total (BDT)	Reference
Mini EV Hubs (Solar + Grid)	15 lakh / hub	20 hubs	3.00 crore	IDCOL EV Program 2024 – Solar Charging Pilot Costs
Battery Swap Units	1 lakh / unit	40 units (2 per hub)	0.40 crore	SREDA EV Roadmap 2022, Local Vendor Pricing
Battery Inventory (BaaS)	25k / battery	500 batteries	1.25 crore	IEA Global EV Outlook 2024 (2W/3W battery cost)
VoltEdge Super App	20 lakh dev + 5 lakh ops	1 yr	0.25 crore	Benchmark from Shohoz & Pathao App Dev Costs
Staff & O&M	10k / hub / month	20 hubs × 12 months	0.25 crore	Local O&M labor benchmark (IDCOL Solar Mini-Grid)
Marketing & Awareness			0.10 crore	Digital & Fleet Campaign Costs (Uber Bangladesh 2023)

Total **5.25 crore BDT (~\$0.43M USD)**



VOLTEGE REVENUE & INVESTOR MODEL



Payback & ROI Trajectory

Year 1-2: Trust-building, low margin

Year 3: Revenue inflection with fleet & corridor adoption

Break-even: Year 4 (~2028)

2027 → 25% cumulative

2028 → 70% cumulative

2029 → 130% cumulative (20% annual ROI)



Revenue Streams & Projections

Year	Users (BaaS)	Total Revenue (Cr BDT)	OPEX (Cr BDT)	Net Profit (Cr BDT)	Cumulative Profit
2025-26	2,000	7.0	6.0	1.0	1.0
2027	5,000	8.5	6.0	2.5	3.5
2028	7,000	10.0	6.5	3.5	7.0
2029	10,000	12.0	8.0	4.0	11.0



Why Investors Will Join

Proven Payback: ~3.5 years via BaaS subscriptions

Carbon Credits: Additional monetization potential

Scalable Model: 500+ hubs & 6,00,000 EVs by 2030

ESG Alignment: Meets SDG 7 & SDG 13 (Clean Energy & Climate Action)



Revenue Breakdown

Waterfall Flow of Revenue (in Pilot Phase)

1. Battery-as-a-Service (BaaS) → **60%** → 4.2 Cr/year
2. Fast Charging → **25%** → 1.8 Cr/year
3. Fleet Dashboard & Data Services → **12%** → 0.8 Cr/year
4. Super App Monetization → **3%** → 0.2 Cr/year

Scaling to **500+ hubs** can raise revenue 5X by 2030.



Target Investors & Partnerships

Equity & PPP Blend:

1. **Govt / Policy** → SREDA, Power Division (land, tax support)
2. **Development & Green Finance** → IDCOL, World Bank, ADB, UNDP-GEF
3. **Private Partners** → Uber, Pathao, Shohoz, Runner, Walton, Rahimafrooz
4. **VC & Impact Funds** → Startup Bangladesh, BD Angels, BlueOrchard, GCF

WHY THIS MODEL WILL WORK IN BANGLADESH

01 **Bangladesh is EV-Ready – Just Needs a Push**

Already **2.5+ lakh** EVs (mostly 2 & 3-wheelers) on road

High urban density = ideal for **micro-hub battery** swap model

65% smartphone penetration → perfect for Super App model

Source: BTRC 2024, IDCOL EV Status Report

02 **Low-Cost Tech for a Low-Income Market**

Modular Solar + Grid hybrid hubs = **cheaper** than fuel stations

Battery-as-a-Service reduces EV **cost by 20-30%**

03 **Digital-First Strategy = Rapid Scaling**

People already use **Pathao, Uber, bKash** → familiar UX

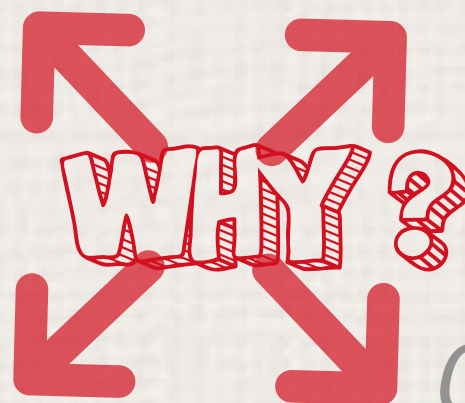
VoltEdge App = real-time battery swap + route + **loyalty rewards**

04 **Massive Environmental ROI**

Each swap = **0.5-1.2 kg** CO₂ saved

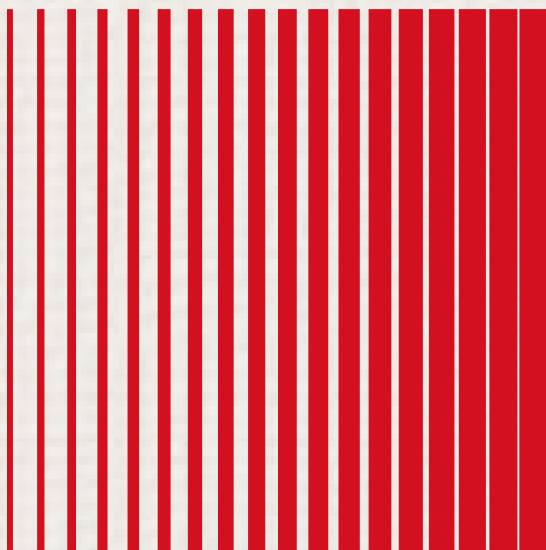
Estimated **0.3 Mt** CO₂ reduction by 2030

Adds to **SDG 7 (Clean Energy)** & **SDG 13 (Climate Action)**



CONTINGENCY *PLAN*

Risk Area	Contingency Strategy Adopted From
Battery Shortage	Maintain 15% extra battery stock at hubs India - Sun Mobility
Power Outage	Solar + genset backup in 50% hubs Vietnam - VinFast
Low User Adoption	Free swaps (3 mo) + lease-to-own battery plans Indonesia - Gojek EV
Policy Delays	Secure city-level PPP with DNCC, CCC India - FAME-II cities
Financial Instability	3-phase rollout + 6-month reserve fund Vietnam - Hub scaling



"Inspired by Real Success in Similar Economies"
(India, Vietnam, Indonesia)





THANK YOU!

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